True or false?

"At most 12" excludes 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 1

Satisfy several constraints at once • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Which whole-number values of x give a volume of at least 500 cm^3 AND a height of at most 3 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

A ramp must rise 2 ft and fit within 20 ft of horizontal space. Can it meet the 1 : 12 limit? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

True or false?

A design that satisfies one constraint may fail another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

A rectangular garden needs area 48 m² and width at most 6 m. Which works?

A. 6 m wide by 8 m long

B. 4 m wide by 10 m long

C. 7 m wide by 7 m long

D. 8 m wide by 6 m long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

A cylinder 3 m tall must hold at least 10 m³ and stand on a 2 m by 2 m floor space. Is that possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 2

Find the best design with a function graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Write its volume as a function V(x) and state the domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

An open box made from a 20 cm by 20 cm sheet with x-cm corner squares cut out has volume $V(x) = x(20 - 2x)^2$. Use a graph or table to estimate the x that gives the largest volume, to the nearest tenth, and that volume to the nearest cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

True or false?

$V(x) = x(20 - 2x)^2$ has zeros at $x = 0$ and $x = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

For an open box made from a 20 cm by 20 cm sheet, $V(x) = x(20 - 2x)^2$. The graph of V on $0 < x < 10$ rises and then falls. Its highest point gives

A. the cut that makes no box

B. the base area

C. the largest volume

D. the smallest volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S86-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

For an open box made from a 20 cm by 20 cm sheet with x-cm corner cuts, $V(x) = x(20 - 2x)^2$ and V is largest near x = 3.3. Why is x = 5 not the best cut even though it is halfway between 0 and 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

SET 87 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to minimize cost and work with typographic grids

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Surface area of prisms and cylinders	**7.G.B.6**
BACKWARD 2	Proportional reasoning with ratios	**7.RP.A.2**
BACKWARD 3	Unit prices and rounding up purchases	**7.EE.B.3**
FORWARD 1	Compare designs with equal volume for least material	**HSG-MG.A.3**
FORWARD 2	Decide between designs using expected cost	**HSS-MD.B.7**

SET 87 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A can with radius 4 cm and height 10 cm is made of metal costing $0.002 per cm². Find the area of metal (top, bottom, and side) and the cost, to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

Can A has r = 4 cm, h = 10 cm. Can B has r = 5 cm, h = 6.4 cm. Show that they hold the same volume, then decide which uses less metal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

A wall 12 m by 3 m has a window 1.5 m by 1.2 m. Paint covers 10 m² per liter and is sold only in whole liters at $8 each. Find the cost to paint the wall once.

Cost: _____

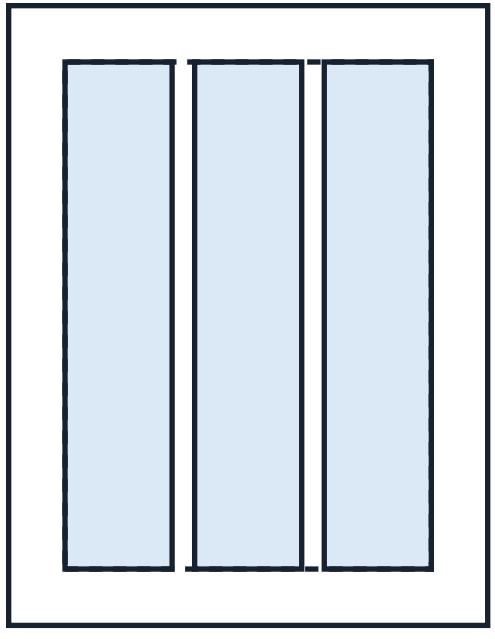
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

A page is 8.5 in. wide with 1 in. margins on each side. The text area has 3 equal columns separated by 2 gutters of 0.2 in. Find the width of each column to the nearest hundredth of an inch.

Width: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

A poster grid uses the golden ratio (about 1.618 : 1) for height : width. A poster is 60 cm wide. Its height is about

A. 120 cm

B. 97.1 cm

C. 37.1 cm

D. 61.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 1

Surface area of prisms and cylinders • Grade 7 • Geometry • 7.G.B.6

Exact area of the two circular ends of a can with radius 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

Exact area of the side of a can with radius 4 and height 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

Surface area of a 2 × 3 × 4 box?

A. 24

B. 48

C. 52

D. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

True or false?

The side of a cylinder unrolls into a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

True or false?

A cylinder's surface area is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 2

Proportional reasoning with ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Height : width = 1.618 : 1 and width = 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q1/5

Split 6.1 in. into 3 equal columns. Width of each (nearest hundredth)?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q2/5

A 2 : 3 rectangle 30 cm wide (width = 2 parts) is

A. 45 cm tall

B. 60 cm tall

C. 20 cm tall

D. 33 cm tall

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q3/5

True or false?

Equal columns divide the available width equally.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q4/5

True or false?

A 1 : 2 ratio means the second is half the first.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 87 • BACKWARD 2 • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 3

Unit prices and rounding up purchases • Grade 7 • Expressions & Equations • 7.EE.B.3

Paint needed is 3.42 L, sold in whole liters. How many liters must be bought?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

Metal costs $0.002 per cm². Cost of 350 cm²?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

Fabric is $6 per yard, sold in whole yards. You need 4.2 yd. Cost?

A. $30

B. $25.20

C. $24

D. $26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

True or false?

When items are sold in whole units, round the amount needed up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

True or false?

3.1 cans of paint can be bought as 3 cans.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

SET 87 • HSG-MG.A.3

FORWARD 1

Compare designs with equal volume for least material • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A box with a square base must hold 1,000 cm³. Compare the surface areas for base sides 8, 10, and 12 cm. Which uses the least material?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

A 12-column grid on a 6.5 in. text area has 11 gutters of 0.125 in. Find the width of each column to the nearest thousandth of an inch.

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

True or false?

Among boxes with a square base and a fixed volume, the cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

A 12-column grid can also be used as

A. only 12 columns

B. 7 equal columns

C. 5 equal columns

D. 4 columns of 3, 3 of 4, or 2 of 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

Why do designers often choose 12-column grids?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

FORWARD 2

Decide between designs using expected cost • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

Packaging A costs $1.20 per item and 2% of items break. Packaging B costs $0.90 and 8% break. A break costs $20. Find the expected cost per item for each and choose.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

A design uses a cheaper metal that fails 5% of the time, with replacement cost $40. How much cheaper must it be per can to be worth it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

True or false?

The cheapest material always gives the cheapest design.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

Expected cost =

A. upfront cost × probability

B. upfront cost + probability of failure × cost of failure

C. cost of failure only

D. upfront cost − probability

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

Name one cost besides material that a can designer might consider.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9 • ADVANCED EXTENSION (+)

Derive and use the formula A = ½ab sin C

MAIN	Geometry main question	**HSG-SRT.D.9**
BACKWARD 1	The sine ratio in a right triangle	**HSG-SRT.C.6**
BACKWARD 2	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 3	The included angle of two sides	**HSG-CO.B.8**
FORWARD 1	Solve for a side or an angle from an area	**HSG-SRT.D.9**
FORWARD 2	From equal areas to the Law of Sines	**HSG-SRT.D.10**

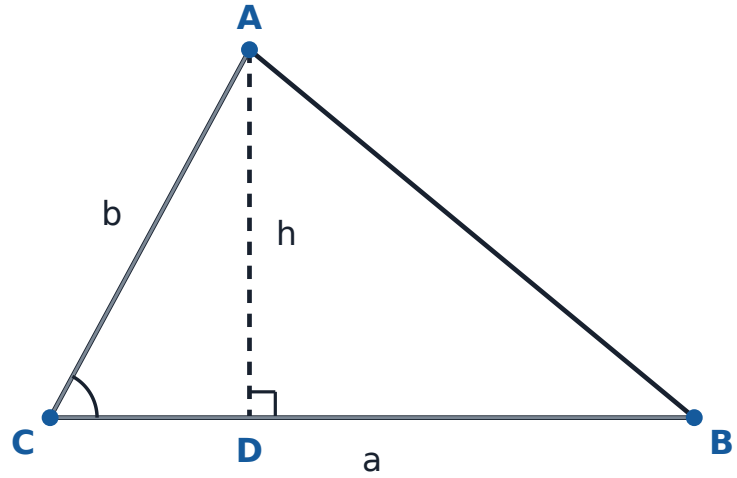
SET 88 • HSG-SRT.D.9

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

In triangle ABC, the altitude from A to side CB has length h. Derive the formula Area = ½ab sin C.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

A triangle has sides a = 8 and b = 11 with included angle C = 40°. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

Two sides of a triangle are 6 and 10, and the angle between them is 150°. Find the exact area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

A parallelogram has sides 7 cm and 12 cm with an angle of 55° between them. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

A triangle has sides 5 and 9 with an included angle of 30°. Its area is

A. 45

B. 22.5

C. 11.25

D. 19.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 1

The sine ratio in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In a right triangle, sin of an acute angle = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

Hypotenuse 10, angle 30°. Find the side opposite the angle.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

If $\sin C = h/b$, then $h =$

A. $b + \sin C$

B. $\sin C/b$

C. $b/\sin C$

D. $b \sin C$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

True or false?

$\sin 90° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

True or false?

$\sin C = \text{adjacent} \div \text{hypotenuse}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 2

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

Base 8, height 5: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

True or false?

Area = ½ · base · height

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5

The height must be ___ to the base.

A. equal

B. adjacent

C. parallel

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

Base a and height b sin C: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

True or false?

A slanted side can be used as the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 3

The included angle of two sides • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In triangle ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

True or false?

SAS uses the angle between the two given sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

For sides a (BC) and b (CA), the included angle is

A. C

B. none

C. A

D. B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

Why must the angle be included for ½ab sin C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

True or false?

Any angle of the triangle works in ½ab sin C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 1

Solve for a side or an angle from an area • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

A triangle has sides 10 and 12 and area 30. Find all possible measures of the included angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

A triangle has area 50, one side 20, and an included angle of 45°. Find the other side to the nearest tenth.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

True or false?

Two different triangles can have the same two sides and the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

The largest area of a triangle with sides 6 and 8 occurs when the included angle is

A. 60°

B. 180°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

Find the area of an equilateral triangle with side 6 using ½ab sin C (exact).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 2

From equal areas to the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

The area of triangle ABC can be written as ½bc sin A, ½ac sin B, or ½ab sin C. Use this to show that sin A/a = sin B/b = sin C/c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

Use sin A/a = sin B/b: A = 30°, a = 5, b = 8. Find sin B.

sin B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

True or false?

The Law of Sines can be derived from the area formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

Dividing ½bc sin A = ½ac sin B by ½abc gives

A. a sin A = b sin B

B. sin A/a = sin B/b

C. sin A = sin B

D. a/b = sin A sin B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

Why is it allowed to divide by ½abc?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Sines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	Use trigonometry in right triangles	**HSG-SRT.C.8**
BACKWARD 2	The area formula ½ab sin C	**HSG-SRT.D.9**
BACKWARD 3	Proportions and cross-multiplying	**7.RP.A.2**
FORWARD 1	Use the Law of Sines (AAS and ASA)	**HSG-SRT.D.10**
FORWARD 2	Surveying with the Law of Sines	**HSG-SRT.D.11**

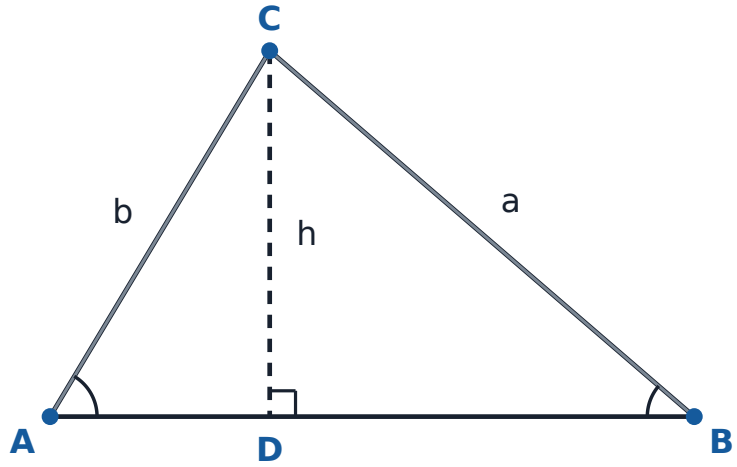
SET 89 • HSG-SRT.D.10

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

In triangle ABC, CD is the altitude to AB. Prove that a/sin A = b/sin B.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

Triangle ABC has an obtuse angle at A, so the altitude from C meets line AB extended past A at D. Prove that a/sin A = b/sin B in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

Explain how the area formula ½bc sin A = ½ac sin B gives the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

In the proof of the Law of Sines, the altitude from C to AB has length h and forms right triangle ADC. Which fact gives $h = b \sin A$?

A. The triangle angle sum

B. $\sin A$ = opposite/hypotenuse in right triangle ADC

C. $\cos A = h/b$

D. The Pythagorean Theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

True or false?

The Law of Sines works only in right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 1

Use trigonometry in right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

Right triangle: hypotenuse 12, angle 40°. Opposite side to the nearest tenth?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

True or false?

An altitude splits a triangle into two right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

If sin B = h/a, then h =

A. h sin B

B. a/sin B

C. a sin B

D. sin B/a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

Right triangle: opposite 5, hypotenuse 10. Angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

True or false?

sin is defined only for angles in right triangles, so it cannot be used after drawing an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 2

The area formula ½ab sin C • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

Triangle ABC has b = 4, c = 5, and A = 90°. What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

True or false?

½bc sin A, ½ac sin B, and ½ab sin C all give the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

Area of a triangle with sides 6, 6 and included 30°?

A. 3

B. 9

C. 36

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

Write the area of triangle ABC using sides a and c.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

True or false?

½ab sin C uses angle A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 3

Proportions and cross-multiplying • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Solve x/6 = 10/4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

True or false?

If b sin A = a sin B, then a/sin A = b/sin B (when the sines are not 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

From 3x = 4y, x/y =

A. 3/4

B. 4/3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

Rewrite $p \cdot q = r \cdot s$ as a proportion with p and r on top.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

True or false?

$a/b = c/d$ means $a + d = b + c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 1

Use the Law of Sines (AAS and ASA) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

In triangle ABC, A = 40°, B = 60°, and a = 10. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

In triangle ABC, A = 40°, B = 60°, and a = 10. Find C, and find c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

True or false?

Knowing two angles and any side determines the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

A = 30°, B = 45°, a = 6. b =

A. $3\sqrt{2} \approx 4.2$

B. $6\sqrt{2} \approx 8.5$

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

Why must the third angle be found before using the Law of Sines for c in an ASA problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 2

Surveying with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

Two observers 500 m apart sight a tree across a river. The angles at the observers (between the baseline and the tree) are 70° and 60°. How far is the tree from the observer at the 70° angle (nearest meter)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

A surveyor measures a baseline of 200 m and angles of 52° and 71° to a peak. Find the distance from the 52° end to the peak (nearest meter).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

True or false?

Surveyors use a measured baseline and two angles to find distances they cannot walk.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

In a surveying triangle, the side opposite the largest angle is

A. equal to the others

B. the shortest

C. the longest

D. the baseline always

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

Explain why measuring angles is easier than measuring the distance across the river directly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Cosines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	The distance formula	**8.G.B.8**
BACKWARD 3	Sine and cosine in a right triangle	**HSG-SRT.C.6**
BACKWARD 4	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Use the Law of Cosines (SAS and SSS)	**HSG-SRT.D.10**
FORWARD 2	Distances between moving objects	**HSG-SRT.D.11**

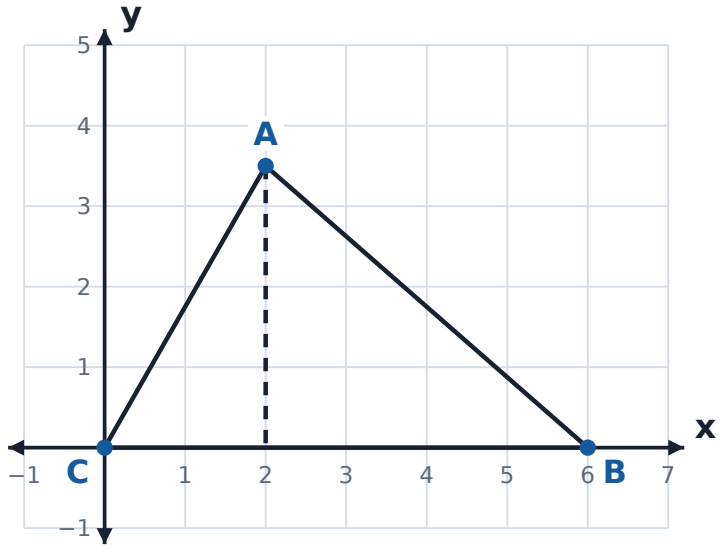
SET 90 • HSG-SRT.D.10

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

Place triangle ABC with C at the origin and B at (a, 0), so A = (b cos C, b sin C). Prove that $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

In acute triangle ABC, the altitude from A meets BC at D. Let CD = x. Use the two right triangles to prove $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

Show that the Law of Cosines becomes the Pythagorean Theorem when C = 90°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

If angle C is obtuse, then

A. $c = a + b$

B. $c^2 = a^2 + b^2$

C. $c^2 < a^2 + b^2$

D. $c^2 > a^2 + b^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

True or false?

The Law of Cosines applies only to acute triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

Legs 5 and 12: hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

Hypotenuse b, leg x: the other leg squared?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

In triangle ABC, the altitude from A to BC has length h and foot D, with CD = x and BC = a. In right triangle ABD, $h^2 + (a - x)^2 =$

A. a^2

B. c^2

C. x^2

D. b^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

True or false?

The Pythagorean Theorem applies to the right triangles made by an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

True or false?

$a^2 + b^2 = c^2$ in every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 2

The distance formula • Grade 8 • Geometry • 8.G.B.8

Distance from (0, 0) to (3, 4)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

Write the squared distance from (p, q) to (a, 0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

Squared distance from (1, 2) to (4, 6)?

A. 5

B. 25

C. 49

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

True or false?

Distance from (2, 0) to (5, 0) is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 3

Sine and cosine in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Hypotenuse b and angle C: adjacent side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

Hypotenuse b and angle C: opposite side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

For a 3-4-5 triangle, $(\sin)^2 + (\cos)^2$ of an acute angle =

A. 25

B. 1

C. 12/25

D. 7/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-B3-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

True or false?

$\sin^2 C + \cos^2 C = 1$ for acute angles C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

True or false?

cos C = opposite ÷ hypotenuse

A. True

B. False

SET 90 • HSG-SRT.D.10

BACKWARD 4

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(b \cos C - a)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q1

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

Expand $(a - x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q2

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

$(p - q)^2 =$

A. $p^2 - 2pq + q^2$

B. $p^2 - pq + q^2$

C. $p^2 - q^2$

D. $p^2 + q^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q3

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

True or false?

$$(a - x)^2 - x^2 = a^2 - 2ax$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q4

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

True or false?

$$(a - x)^2 = a^2 - x^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q5

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 1

Use the Law of Cosines (SAS and SSS) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

In triangle ABC, a = 7, b = 9, and C = 50°. Find c to the nearest tenth.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

A triangle has sides 5, 7, and 8. Find the angle opposite the side of length 7.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

True or false?

The Law of Cosines can find an angle when all three sides are known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

Sides 3, 4, 6. The angle opposite 6 is

A. obtuse

B. acute

C. right

D. impossible to tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

Find the largest angle of the triangle with sides 4, 5, and 6, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 2

Distances between moving objects • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

Two ships leave port at the same time on courses 70° apart. One travels 20 km and the other 30 km. How far apart are they, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

A golfer's ball is 150 yd from the tee; the hole is 180 yd from the tee; the angle between the two directions is 12°. How far is the ball from the hole (nearest yard)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

True or false?

The Law of Cosines can be used only when the angle between the two known sides is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

Two hikers walk 3 km and 5 km from camp in directions 60° apart. Distance between them?

A. 8 km

B. √34 ≈ 5.8 km

C. 2 km

D. √19 ≈ 4.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

Two ships leave port on courses 70° apart. Why can't the Pythagorean Theorem alone give the distance between them?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Sines, including the ambiguous case

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Sines	**HSG-SRT.D.10**
BACKWARD 2	Inverse sine	**HSG-SRT.C.8**
BACKWARD 3	Triangle angle sum	**HSG-CO.C.10**
BACKWARD 4	Sine of supplementary angles	**HSG-SRT.D.9**
FORWARD 1	Solve both triangles in the ambiguous case	**HSG-SRT.D.11**
FORWARD 2	Bearings and navigation with the Law of Sines	**HSG-MG.A.1**

SET 91 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

Points A and B are 120 m apart on one bank of a river. A tree C is on the other bank. $\angle A = 72°$ and $\angle B = 63°$. Find AC to the nearest tenth of a meter.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

In triangle ABC, A = 30°, a = 6, and b = 10. How many triangles are possible? Find all possible measures of B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

In triangle ABC, A = 40°, a = 12, and b = 10. How many triangles are possible? Find B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

In triangle ABC, A = 50°, a = 5, and b = 10. How many triangles are possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

A is acute and $b \sin A < a < b$. How many triangles have these measurements?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 1

The Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

Write the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

A = 30°, B = 90°, a = 5. b =

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

Solving a/sin A = b/sin B for sin B gives

A. b sin A/a

B. a/(b sin A)

C. sin A

D. a sin A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

True or false?

Each side in the Law of Sines is paired with its opposite angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

True or false?

$a/\sin B = b/\sin A$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 2

Inverse sine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

sin B = 0.5, B acute. B =

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

sin B = 0.8. Find the acute B to the nearest tenth.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

$\sin^{-1}(1) =$

A. 0°

B. 180°

C. 45°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B2-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

True or false?

The calculator's $\sin^{-1}$ always gives an angle between $-90°$ and $90°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

True or false?

$\sin^{-1}(1.2)$ is an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 3

Triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Two angles are 72° and 63°. Third angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

True or false?

A triangle cannot have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

A = 40° and B = 147.6°. Is this a triangle?

A. Only if C is 0°

B. No: the sum exceeds 180°

C. Yes

D. Only if right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

A = 30°, B = 123.6°. Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

True or false?

The angles of a triangle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 4

Sine of supplementary angles • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

True or false?

$\sin 150° = \sin 30°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

Name an obtuse angle with the same sine as 56.4°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

sin 120° equals

A. cos 120°

B. sin 60°

C. sin 30°

D. −sin 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

Why might sin B = 0.83 give two possible triangles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

True or false?

An obtuse angle has a negative sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 1

Solve both triangles in the ambiguous case • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

A = 30°, a = 6, b = 10. For the triangle with acute B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

In triangle ABC, A = 30°, a = 6, and b = 10, so B ≈ 56.4° or B ≈ 123.6°. For the triangle with obtuse B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

True or false?

In the ambiguous case the two triangles have different third sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

A = 30°, b = 10, and a = 5. How many triangles?

A. Three

B. Two

C. One (a right triangle)

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

A = 120°, a = 8, b = 10. How many triangles? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 2

Bearings and navigation with the Law of Sines • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

A lighthouse is spotted from a ship at an angle of 35° to its course. After the ship sails 8 km farther, the angle is 70°. How far is the ship from the lighthouse at the second sighting (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

A forest ranger at A sees a fire at an angle of 48° from the line to a second tower B, 10 km away. From B, the angle is 61°. How far is the fire from A (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

True or false?

Bearings and sightings are modeled as triangles with a known side and two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

A ship sights a lighthouse at 35° to its course; after sailing 8 km farther along the course, the angle is 70°. In the triangle formed by the two ship positions and the lighthouse, the angle at the lighthouse is

A. 75°

B. 110°

C. 70°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

Name one reason a navigation model using a flat triangle might be inaccurate over very long distances.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Cosines, including resultant forces

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Cosines	**HSG-SRT.D.10**
BACKWARD 2	Inverse cosine	**HSG-SRT.C.8**
BACKWARD 3	Angles in a parallelogram	**HSG-CO.C.11**
FORWARD 1	Solve a triangle with both laws	**HSG-SRT.D.11**
FORWARD 2	Velocity vectors: airspeed and wind	**HSN-VM.A.3**

SET 92 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

To find the length AB across a lake, a surveyor at C measures CA = 350 m, CB = 420 m, and ∠C = 68°. Find AB to the nearest tenth of a meter.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

Two forces of 40 N and 60 N act on an object with an angle of 50° between them. Find the magnitude of the resultant force to the nearest tenth.

Resultant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

A triangle has sides 8, 10, and 13. Find its largest angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

A plane flies 200 km, turns 25° to the right, and flies 150 km. How far is it from its start, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

Which law should be used first to solve a triangle given all three sides?

A. The Law of Sines

B. The Pythagorean Theorem

C. The angle sum

D. The Law of Cosines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 1

The Law of Cosines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

Write the Law of Cosines for side c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

a = 3, b = 4, C = 90°: c?

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

Solved for cos C:

A. $(a^2 + b^2 - c^2)/(2ab)$

B. $(c^2 - a^2 - b^2)/(2ab)$

C. $2ab/(a^2 + b^2 - c^2)$

D. $(a^2 + b^2 + c^2)/(2ab)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-B1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

True or false?

The angle in the formula is the one opposite the side being found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q4/5

True or false?

$$c^2 = a^2 + b^2 + 2ab \cos C$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 2

Inverse cosine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

$\cos C = 0.5$. $C =$

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

cos C = −0.03125. C to the nearest tenth?

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

A negative cosine means the angle is

A. acute

B. right

C. obtuse

D. zero

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

True or false?

$\cos^{-1}$ gives angles from 0° to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

True or false?

$\cos^{-1}(0) = 0°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 3

Angles in a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

One angle of a parallelogram is 50°. The angle next to it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

Forces at 50° form a parallelogram; the angle opposite the resultant in the triangle is

A. 130°

B. 25°

C. 50°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

A plane turns 25°. The angle inside the triangle at the turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

True or false?

The angle between two forces is the angle used in the Law of Cosines for the resultant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

SET 92 • HSG-SRT.D.11

FORWARD 1

Solve a triangle with both laws • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

In triangle ABC, b = 12, c = 15, A = 40°. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

In triangle ABC, b = 12, c = 15, and A = 40°. Find B to the nearest tenth of a degree (use the Law of Sines or Cosines).

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

True or false?

After the Law of Cosines, it is safest to find the smaller remaining angle with the Law of Sines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

Given two sides and the included angle, use first

A. $\sin^{-1}$

B. the Law of Sines

C. the Law of Cosines

D. the angle sum

Use Heron's formula or ½ab sin C to find the area of the triangle with sides 8, 10, 13 (nearest tenth).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

FORWARD 2

Velocity vectors: airspeed and wind • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

A plane flies due east at 300 km/h; a wind blows due north at 50 km/h. Find its ground speed (nearest tenth) and direction (degrees north of east, nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

A boat heads across a river at 8 km/h; the current is 3 km/h at 90° to its heading. Find its actual speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

True or false?

The resultant of two velocities is always found by adding their speeds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

Velocities of 10 and 10 with 120° between them have a resultant of

A. 0

B. 17.3

C. 10

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

A plane's velocity is 400 km/h; a 60 km/h wind blows at 45° to its heading. Find the ground speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

SET 93 • HSG-C.A.4 • ADVANCED EXTENSION (+)

Construct a tangent line from a point outside a circle

MAIN	Geometry main question	**HSG-C.A.4**
BACKWARD 1	Construct the midpoint of a segment	**HSG-CO.D.12**
BACKWARD 2	Inscribed angles on a diameter	**HSG-C.A.2**
BACKWARD 3	Tangent is perpendicular to the radius	**HSG-C.A.2**
FORWARD 1	Prove that the two tangent segments are congruent and compute…	**HSG-C.A.4**
FORWARD 2	Lines of sight to the horizon	**HSG-MG.A.1**

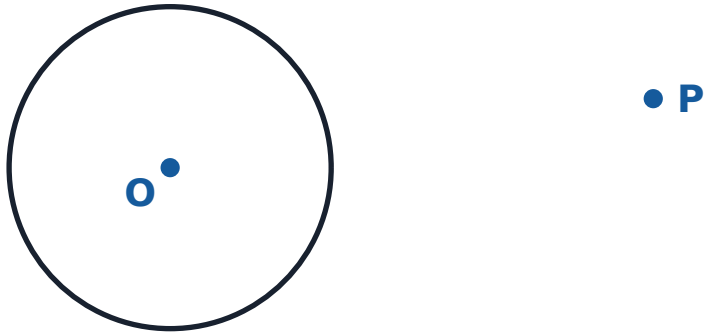
SET 93 • HSG-C.A.4

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

Compass and straightedge only; leave all arcs.

Construct a line through P tangent to the circle with center O.



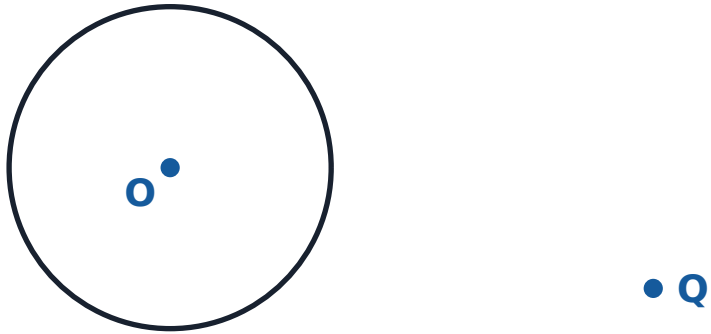
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct BOTH tangent lines from Q to the circle with center O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

Write the steps for constructing a tangent from an external point P to circle O.

Steps: _____

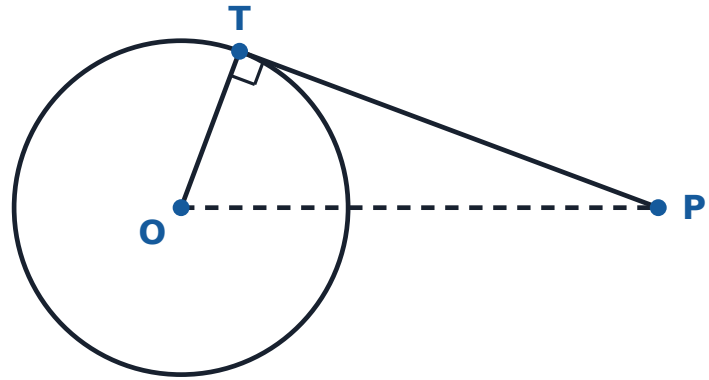
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

Explain why line PT in the construction is tangent to circle O.

Work space



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

The two tangent segments from P to circle O are

A. perpendicular

B. congruent

C. parallel

D. always equal to the radius

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 1

Construct the midpoint of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction finds the midpoint of OP?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector crosses the segment at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

The perpendicular bisector construction uses arcs

A. of any radius

B. centered at the midpoint

C. of equal radius from both endpoints

D. from one endpoint only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

To construct a tangent from a point P outside circle O, why is the midpoint of OP needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

True or false?

The midpoint can be found by eye in a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 2

Inscribed angles on a diameter • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

An angle inscribed in a semicircle is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

A point T is on the circle with diameter OP. m∠OTP = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

The intercepted arc of an angle inscribed in a semicircle is

A. 180°

B. 360°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, why is a circle with diameter OP drawn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

True or false?

An inscribed angle on a diameter can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 3

Tangent is perpendicular to the radius • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

A tangent meets the radius at what angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

If PT $\perp$ OT and T is on circle O, then PT is

A. a secant

B. a diameter

C. a chord

D. tangent to the circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

OT = 5 and OP = 13 with PT tangent at T. PT = ?

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

True or false?

A tangent passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 1

Prove that the two tangent segments are congruent and compute lengths • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

PA and PB are tangent to circle O at A and B. Prove that PA = PB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

A circle has radius 5 and P is 13 from its center. Find the length of each tangent segment from P.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

True or false?

OP bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

Tangents from P make a 60° angle. The angle AOB is

A. 30°

B. 120°

C. 60°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

Radius 6, tangent segment 8. Distance from P to the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 2

Lines of sight to the horizon • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

Model Earth as a sphere of radius 6,371 km. A person's eyes are 2 m (0.002 km) above the sea. How far is the horizon (tangent distance), to the nearest tenth of a km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

From the top of a 100 m tower (0.1 km), how far is the horizon, to the nearest km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

True or false?

The line of sight to the horizon passes through Earth's center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

Raising the eye height by a factor of 4 multiplies the horizon distance by about

A. 16

B. 8

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

In the model of the distance to the horizon from a point above Earth, why is the right angle of the triangle at the horizon point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of an ellipse from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive a circle equation	**HSG-GPE.A.1**
BACKWARD 3	Expand squares	**HSA-APR.A.1**
FORWARD 1	Read foci and axes from an ellipse equation	**HSG-GPE.A.3**
FORWARD 2	Design with the reflective property of ellipses	**HSG-MG.A.3**

SET 94 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

An ellipse is the set of points whose distances to the foci $(-3, 0)$ and $(3, 0)$ add to 10. Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

Write the equation of the ellipse with foci (0, −4) and (0, 4) whose distances to the foci add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

Write the equation of the ellipse with foci (±5, 0) and vertices (±13, 0).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

Write the equation of the ellipse with foci (1, 2) and (7, 2) whose distances to the foci add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

For any point on $x^2/36 + y^2/20 = 1$, the sum of its distances to the foci is

A. 12

B. $2\sqrt{20}$

C. 36

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

Distance from (x, y) to (3, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

Distance from (0, 4) to (3, 0)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

Distance from (5, 0) to (−3, 0)?

A. √34

B. 64

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

True or false?

The distance formula has a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

True or false?

$$\sqrt{(a^2 + b^2)} = a + b$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 2

Derive a circle equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Equation of the circle with center (0, 0) and radius 5?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

True or false?

A circle is an ellipse whose two foci coincide.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

$x^2/25 + y^2/25 = 1$ is

A. an ellipse with different axes

B. a parabola

C. a circle of radius 5

D. a hyperbola

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

A circle is the set of points at a fixed distance from

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

True or false?

An ellipse has one focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 3

Expand squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(x + 3)^2 - (x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

$(3x - 25)^2 =$

A. $9x^2 + 625$

B. $9x^2 - 75x + 625$

C. $3x^2 - 150x + 625$

D. $9x^2 - 150x + 625$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

True or false?

$$(a - b)^2 = a^2 - 2ab + b^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 1

Read foci and axes from an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Find the foci of $x^2/25 + y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

Find the vertices and foci of $x^2/16 + y^2/25 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

True or false?

For an ellipse, $c^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

Which ellipse is closest to a circle?

A. $x^2/100 + y^2/9 = 1$

B. $x^2/25 + y^2/4 = 1$

C. $x^2/25 + y^2/24 = 1$

D. $x^2/25 + y^2/1 = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

Find the length of the major axis of $x^2/49 + y^2/36 = 1$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 2

Design with the reflective property of ellipses • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A whispering gallery is an ellipse 40 m long and 16 m wide. How far from the center should two people stand (at the foci), to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

A medical device places a kidney stone at one focus of an ellipse $x^2/100 + y^2/64 = 1$ (cm). Where is the emitter placed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

True or false?

Sound from one focus of an elliptical room reflects back to the same focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

A garden ellipse is drawn with a 10 m rope tied to stakes 6 m apart. Its width (minor axis) is

A. 8 m

B. 10 m

C. 4 m

D. 6 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

Explain how the string-and-stakes method uses the definition of an ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of a hyperbola from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive an ellipse equation	**HSG-GPE.A.3**
BACKWARD 3	Absolute value as distance	**6.NS.C.7**
FORWARD 1	Read foci and asymptotes from a hyperbola equation	**HSG-GPE.A.3**
FORWARD 2	Locate a position from a time difference	**HSG-MG.A.3**

SET 95 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

A hyperbola is the set of points whose distances to the foci (−5, 0) and (5, 0) differ by 6 (in absolute value). Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

Write the equation of the hyperbola with foci (0, ±13) whose distances to the foci differ by 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

Write the equation of the hyperbola with foci (±10, 0) and vertices (±6, 0).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

Find the asymptotes of $x^2/9 - y^2/16 = 1$.

Asymptotes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

For any point on $x^2/16 - y^2/9 = 1$, the difference of its distances to the foci is

A. 8

B. 7

C. 10

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

Distance from (x, y) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

Distance from (3, 0) to (5, 0) minus distance from (3, 0) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

Distance from (6, 0) to (−10, 0)?

A. 4

B. 60

C. 16

D. −16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

True or false?

Distances are never negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

True or false?

Distance from (0, 13) to (0, −13) is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 2

Derive an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

True or false?

An ellipse uses a constant SUM of distances to the foci.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

Ellipse with foci (±3, 0) and sum 10: equation?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

For an ellipse, $b^2 =$

A. ac

B. $a^2 - c^2$

C. $a^2 + c^2$

D. $c^2 - a^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

What changes in the definition to get a hyperbola?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

True or false?

An ellipse equation has a minus sign between the squared terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 3

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

$|4 - 10| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

$|10 - 4| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

$|a - b|$ means

A. the size of the difference

B. $a + b$

C. a negative number

D. always $a - b$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

True or false?

$|x| = 6$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

True or false?

$|-3| = -3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 1

Read foci and asymptotes from a hyperbola equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Find the foci of $x^2/16 - y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

Find the vertices and asymptotes of $y^2/4 - x^2/9 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

True or false?

For a hyperbola, $c^2 = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

Which equation is a hyperbola?

A. $x^2/4 + y^2/9 = 1$

B. $y = x^2/4$

C. $x^2/4 - y^2/9 = 1$

D. $x^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

Explain how to tell an ellipse from a hyperbola by its equation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 2

Locate a position from a time difference • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

Two radio stations are 100 km apart at (±50, 0). A ship receives signals showing it is 60 km closer to one station than the other. Write the equation of the hyperbola it lies on.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

Sound travels 0.34 km/s. Two microphones 2 km apart hear a sound 3 s apart. What is the difference in distances to the source?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

True or false?

A single time difference places the source on one branch of a hyperbola, not at a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

Radio stations are at $(-50, 0)$ and $(50, 0)$ (km). A ship on the x-axis, to the right of the origin, is 60 km closer to the right station than to the left one. Its position is

A. $(30, 0)$

B. $(40, 0)$

C. $(50, 0)$

D. $(60, 0)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

Why do navigation systems use at least two pairs of stations?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2 • ADVANCED EXTENSION (+)

Use Cavalieri's principle to justify the volume of a sphere

MAIN	Geometry main question	**HSG-GMD.A.2**
BACKWARD 1	Cavalieri's principle for prisms and cylinders	**HSG-GMD.A.1**
BACKWARD 2	Cross-section radius with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Volumes of cylinders and cones	**HSG-GMD.A.3**
BACKWARD 4	Area of a ring	**7.G.B.4**
FORWARD 1	Use Cavalieri's principle for other solids	**HSG-GMD.A.2**
FORWARD 2	Model a dome as a hemispherical shell	**HSG-MG.A.2**

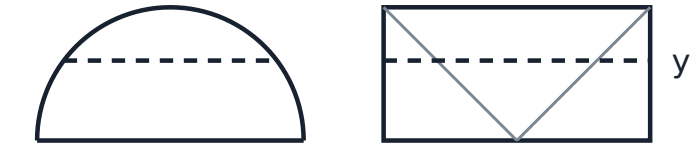
SET 96 • HSG-GMD.A.2

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

A hemisphere of radius r and a cylinder of radius r and height r with a cone removed (as shown) sit on the same plane. Use Cavalieri's principle to show that the hemisphere's volume is $\frac{2}{3}\pi r^3$, and conclude that a sphere has volume $4/3\ \pi r^3$.

Work space



hemisphere, radius r cylinder minus cone

Explanation: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

Let r = 5 and y = 3. Find the cross-section areas of the hemisphere and of the cylinder-minus-cone at height y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

Let r = 6. Find the volume of the cylinder-minus-cone and compare it with $\frac{2}{3}\pi r^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

Explain why a stack of coins pushed into a slanted stack keeps its volume, using Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

Cavalieri's principle requires

A. equal surface areas

B. congruent bases only

C. the same shape at every level

D. equal heights and equal cross-section areas at every level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 1

Cavalieri's principle for prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Solids with equal heights and equal cross-section areas at every level have equal volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

An oblique cylinder has radius 2 and height 5. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

A prism and a cylinder both have base area 30 and height 4. Volumes?

A. Cannot tell

B. 30 and 4

C. 120 and 40

D. Both 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

State what must be equal for Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

True or false?

Cavalieri's principle needs the cross-sections to be the same shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 2

Cross-section radius with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A sphere of radius 5 is sliced 3 units from its center. Radius of the slice?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

Radius r, height y: slice radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

Radius 10, height 6: slice radius

A. 4

B. 8

C. 16

D. $\sqrt{136}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

True or false?

The slice radius, the height, and the sphere radius form a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

True or false?

Slices of a hemisphere get bigger as y increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 3

Volumes of cylinders and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Cylinder r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

Cone r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

Cylinder minus cone (same r and h = r) =

A. $4/3\ \pi r^3$

B. πr^3

C. $\frac{2}{3}\pi r^3$

D. $\frac{1}{3}\pi r^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

True or false?

A cone is ⅓ of the cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q4/5

True or false?

$\pi r^3 - \frac{1}{3}\pi r^3 = \frac{1}{3}\pi r^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 4

Area of a ring • Grade 7 • Geometry • 7.G.B.4

Ring with outer radius 5 and inner radius 3: exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

True or false?

Ring area = $\pi(R^2 - r^2)$

A. True

B. False

Ring with outer 4 and inner 1:

A. 3π

B. 15π

C. 9π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

Ring with outer r and inner y: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

True or false?

Ring area = $\pi(R - r)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 1

Use Cavalieri's principle for other solids • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

A twisted tower and a straight prism both have height 50 m and every horizontal slice is a 10 m by 10 m square. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

A pyramid with base area 36 and height 10 and a cone with base area 36 and height 10: compare volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle shows that every pyramid with base area B and height h has volume ⅓Bh, once one such pyramid is known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

Two solids of height 8 have slice areas 5 and 5 at every level. Volumes?

A. 40 and 13.3

B. 5 each

C. Both 40

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

Two solids have equal heights, but at one level their slice areas differ. Can Cavalieri's principle be used? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 2

Model a dome as a hemispherical shell • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Find the volume of concrete to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

Concrete has density 2.4 t/m³. Estimate the dome's mass to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

True or false?

A shell's volume is the outer volume minus the inner volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

A hemisphere of radius 3 has volume

A. 9π

B. 27π

C. 36π

D. 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

A hemispherical shell has outer radius 10 m and inner radius 9.5 m. Why is its volume NOT $\frac{2}{3}\pi(10 - 9.5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5